



Cont. Agents Architectures + MCP

Week 7

Last Week

Week 4: Tools for building AI Agents

LangChain, Azure Open AI Api, LLMMod.ai

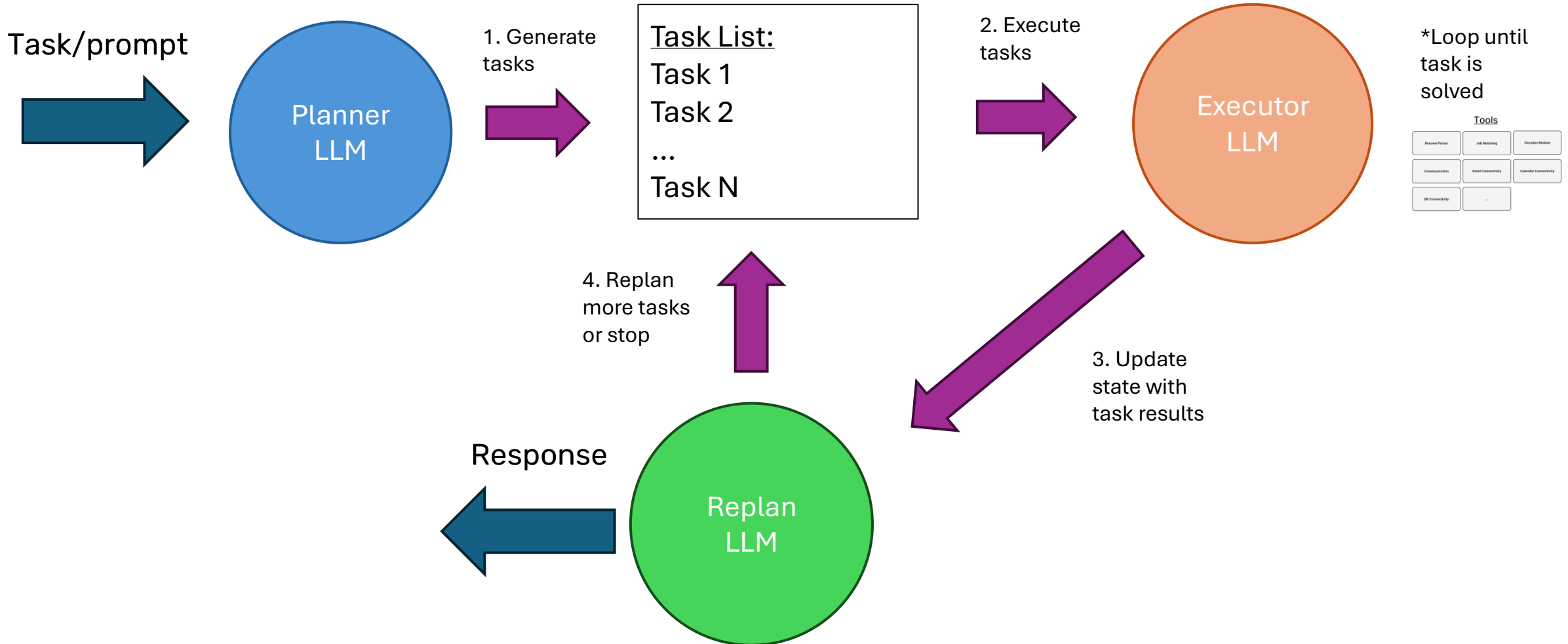
Week 5: AI Workflow & ReAct Agent

AI Workflow, ReAct Agent, Individual Project Launch

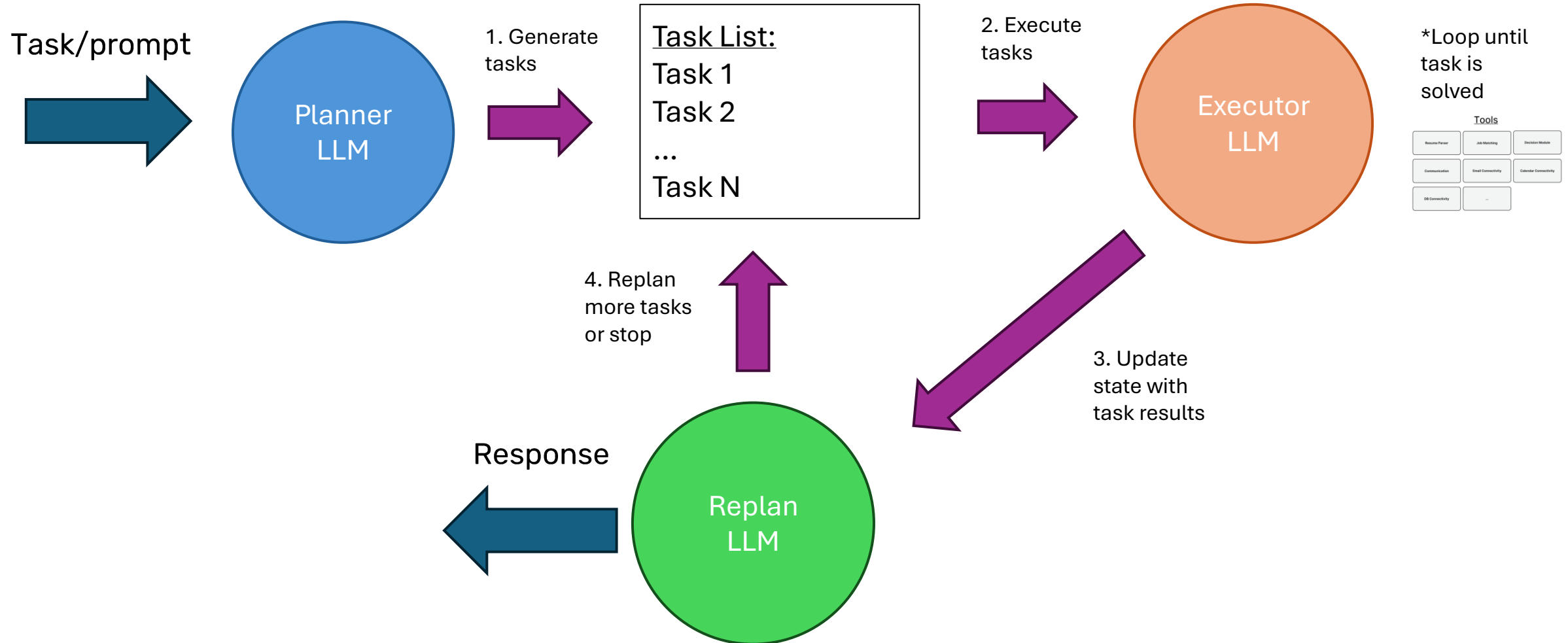
Week 6: Agents Architectures

Cont. ReAct agent, AI Agents Architectures

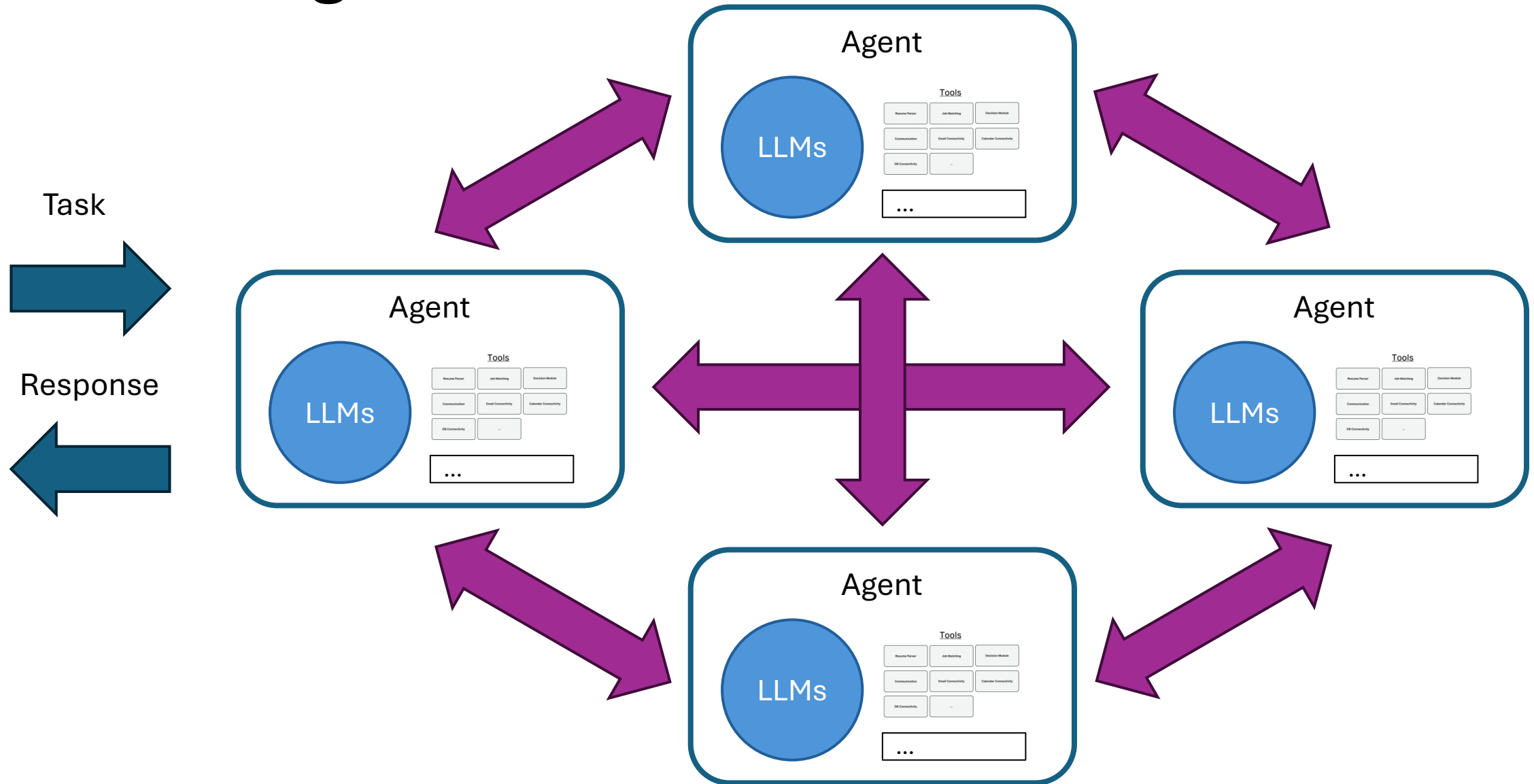
Plan and Execute Agent



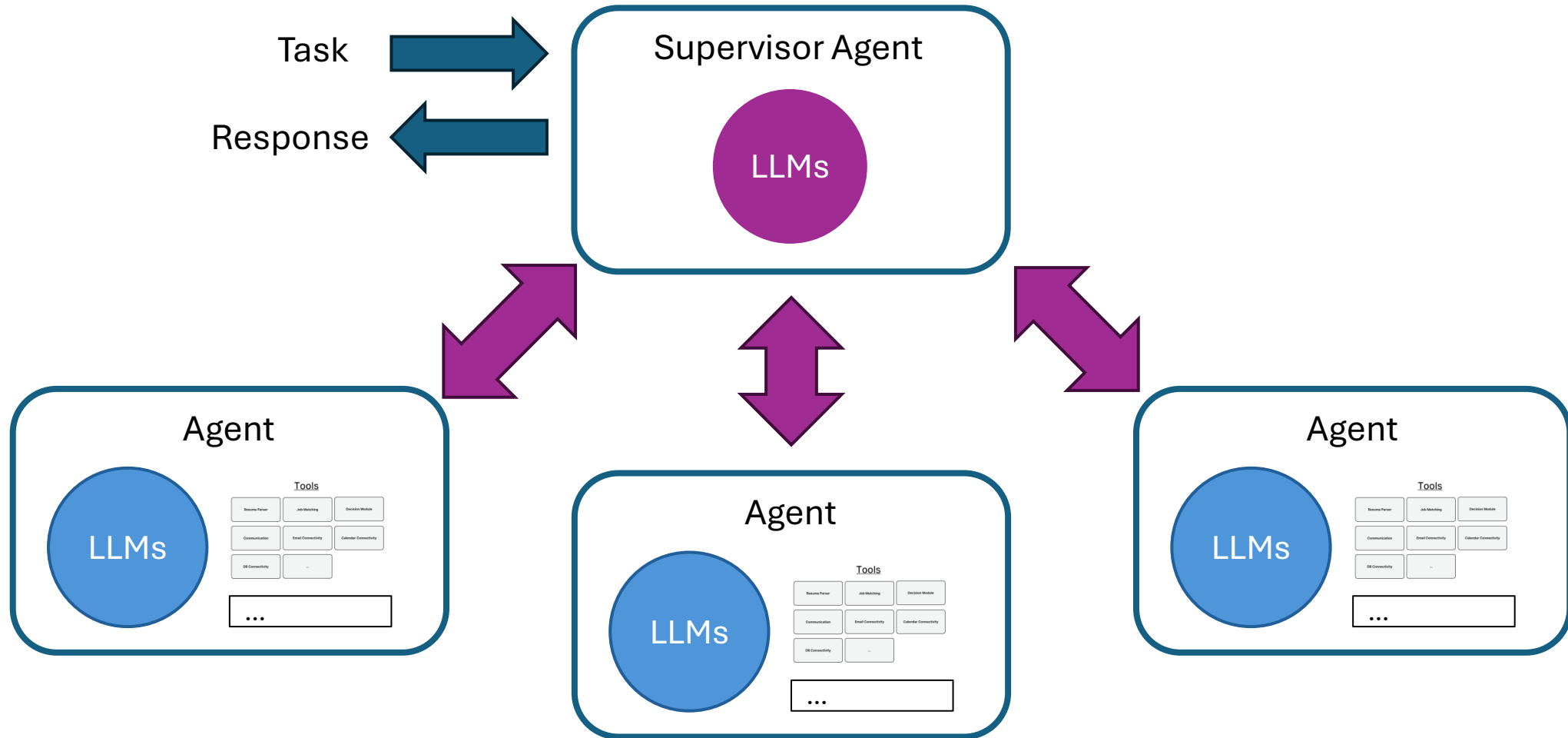
Plan and Execute Agent



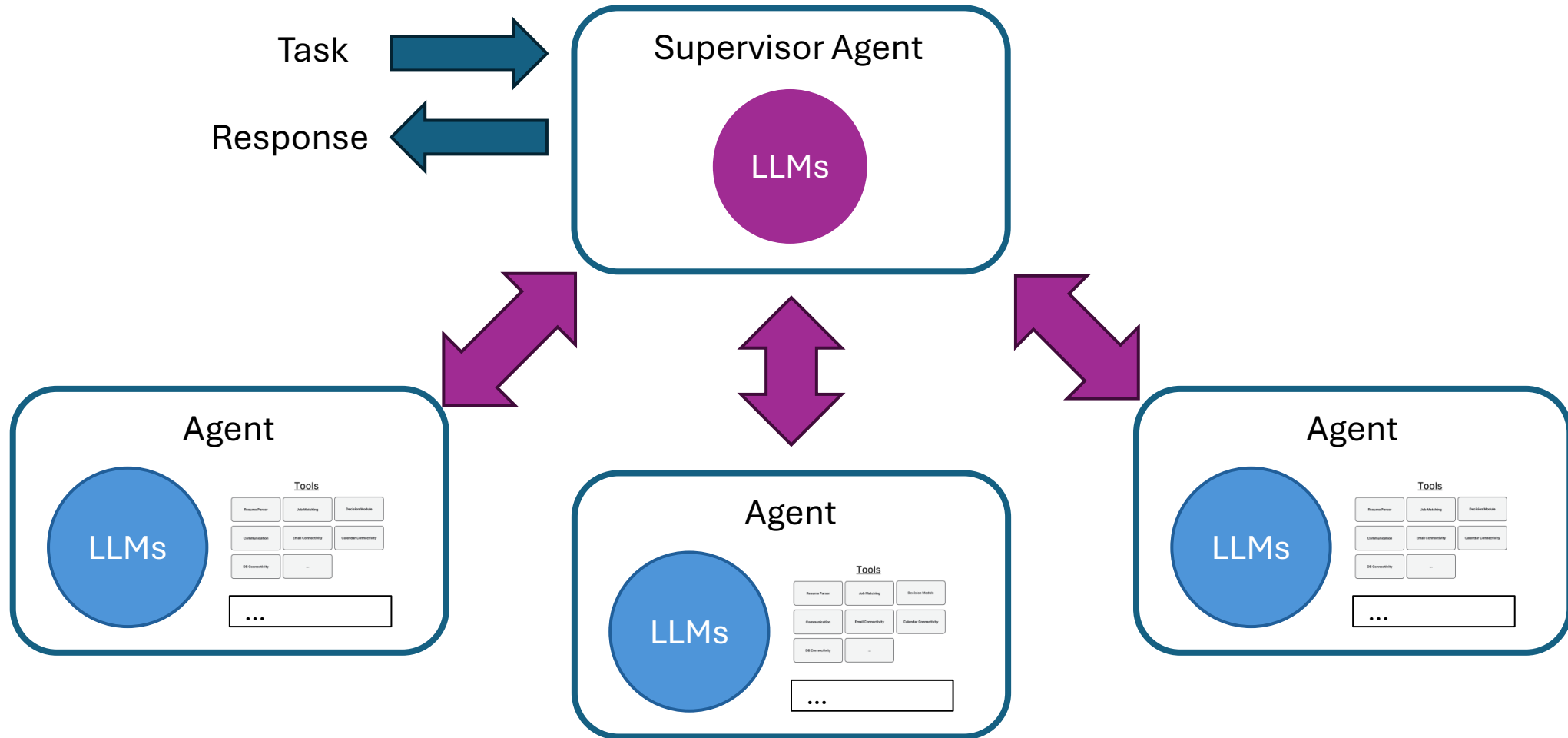
Multi AI Agents



Supervisor Agent



Supervisor Agent



Supervisor Agent

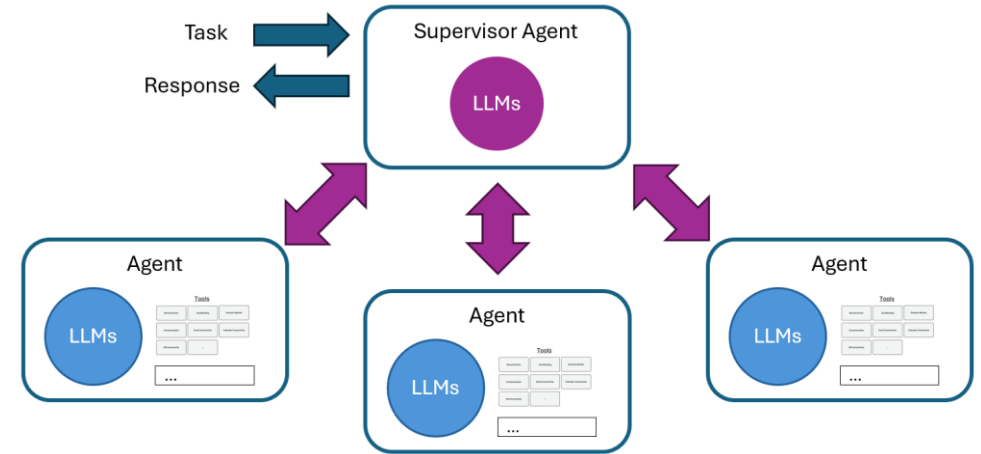
A Supervisor Agent oversees and coordinates other agents. It assigns tasks, monitors progress, resolves conflicts, and decides when to stop or reroute execution.

Good for

Large agent systems and workflows

Safety-critical or high-stakes tasks

Long-running or multi-stage processes



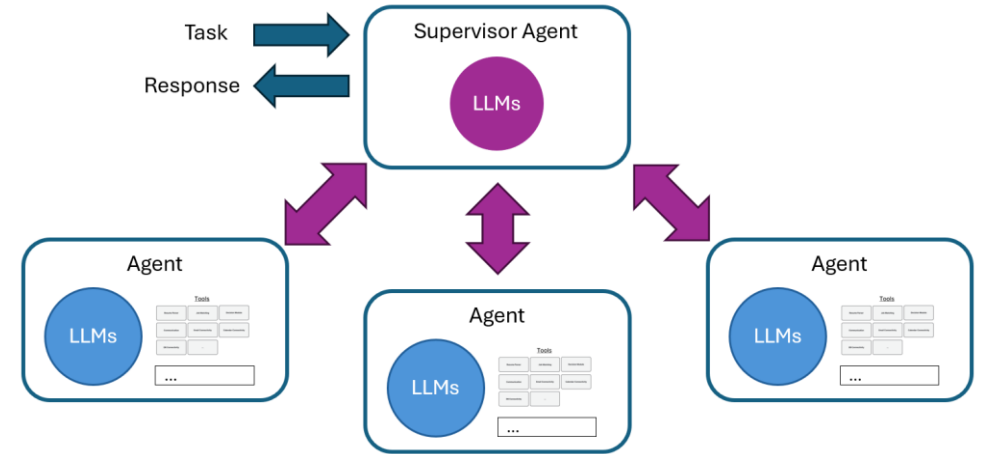
Supervisor Agent

Pros

- Centralized control and governance
- Improves reliability and alignment
- Enables scalable agent architectures

Cons

- Single point of failure
- Additional latency and design complexity
- Requires careful prompt and policy design



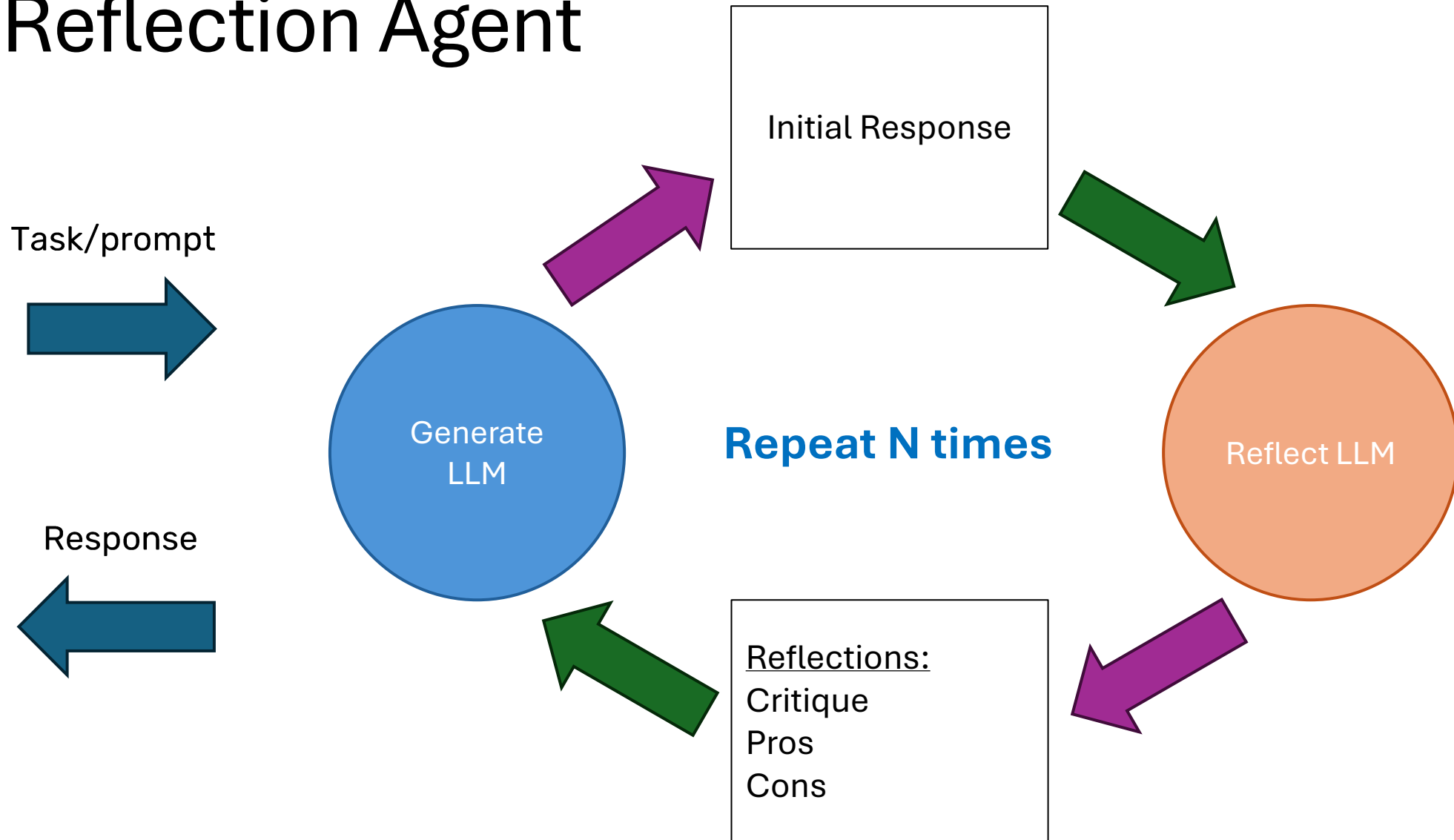
This Week

Week 7: Agents Architectures

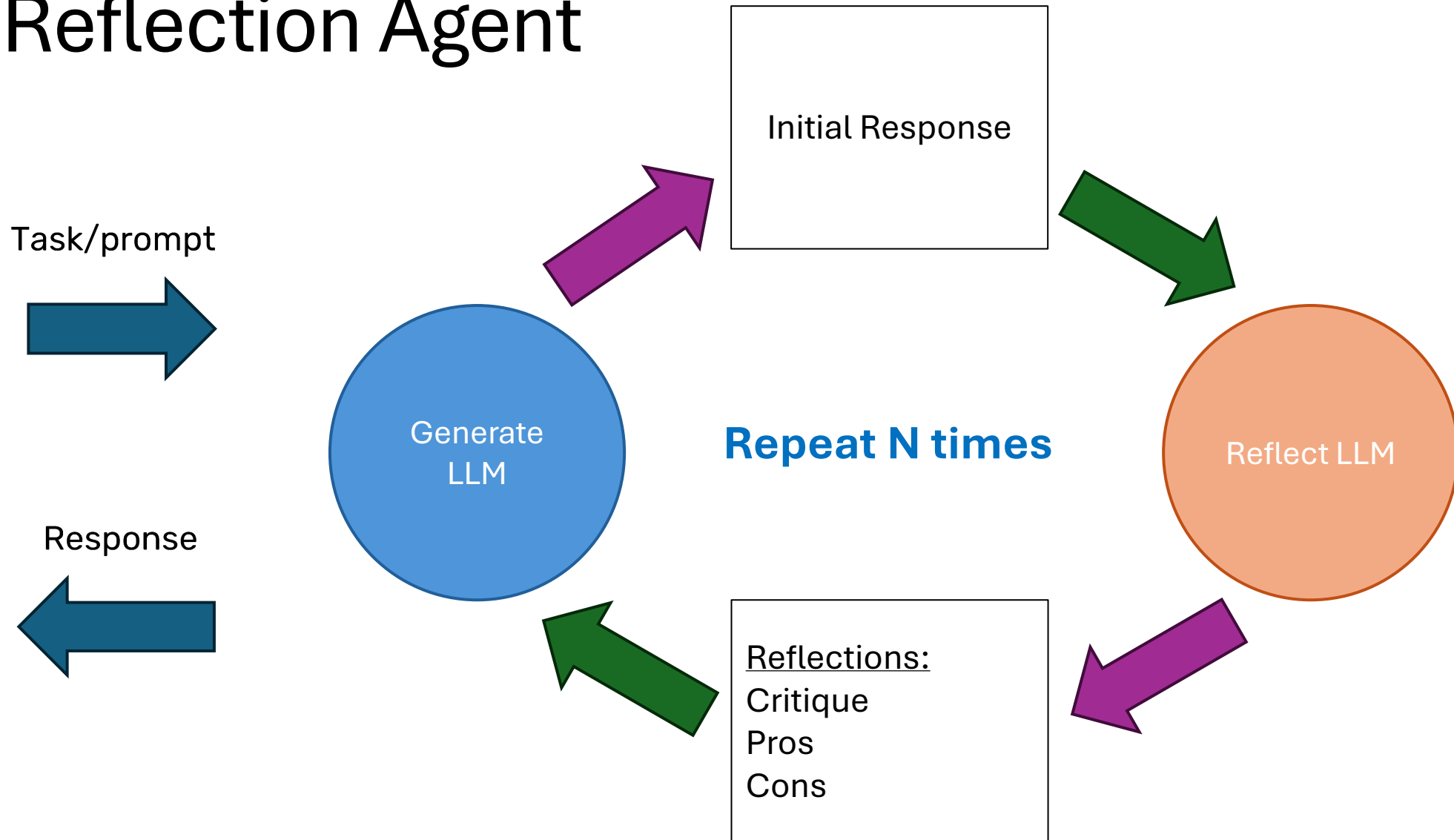
Cont. AI Agents Architectures, MCP

Reflection Agent

Reflection Agent



Reflection Agent



Reflection Agent

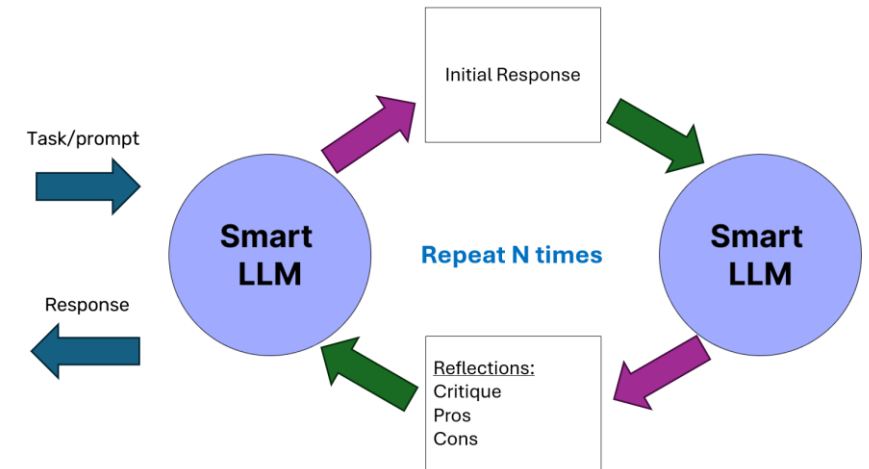
Reflection is a prompting strategy used to improve the quality and success rate of agents and similar AI systems. It involves prompting an LLM to reflect on and critique its past actions, sometimes incorporating additional external information such as tool observations.

Good for

Complex reasoning tasks (Math problems, logic puzzles)

Writing and content refinement

Decision-making under uncertainty



Reflection Agent

Pros

Improves output quality over iterations

Reduces logical errors and hallucinations

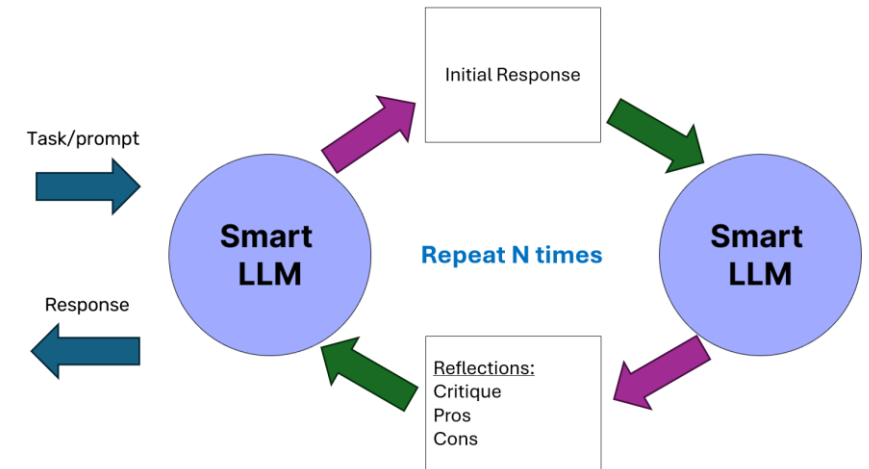
Encourages self-correction and learning from mistakes

Cons

Higher latency and token cost

Can over-reflect and stall progress

Limited benefit for simple or time-critical tasks



MCP

What Does LLM Lacks?



External Context (documents, DB, etc.)



Tools (Search, mail, API, etc.)

MCP: Model Context Protocol

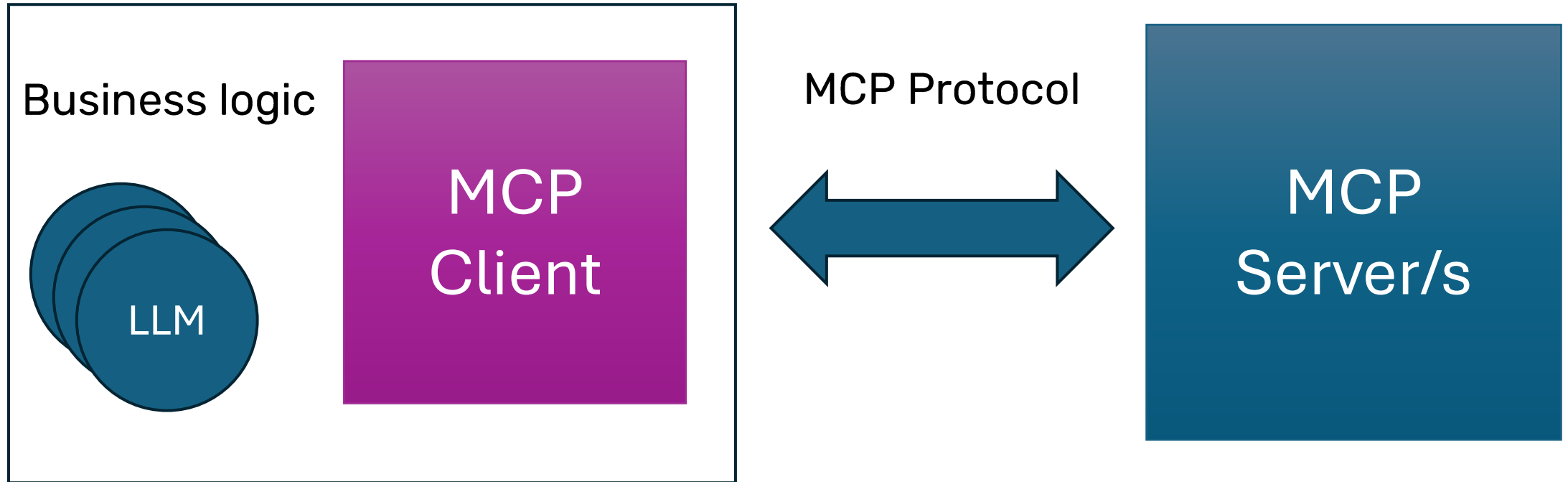
What it is:

The Model Context Protocol (MCP) is an open standard by Anthropic that lets developers securely connect LLMs to external data and tools. Developers can expose data, tools, and reusable prompts through an MCP server, or build AI applications (MCP clients) that consume those capabilities.

Like USB-C (or API*) for AI Apps, allow to connect external tools and context to AI applications in a open standard

How MCP Works

AI Application

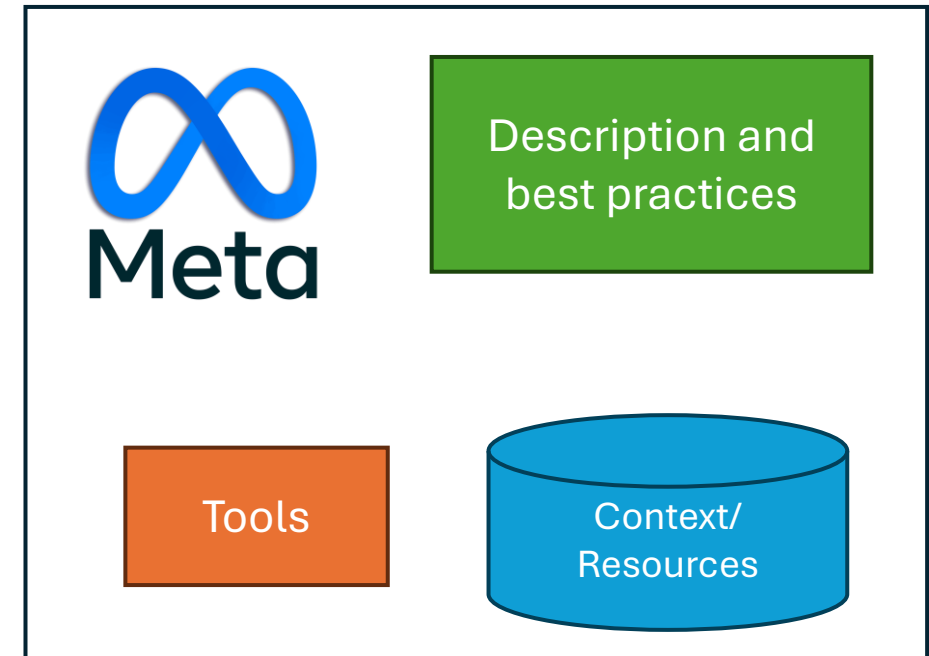


Example - “Meta MCP”

AI App for Social
Marketing



MCP Server



The Protocol

MCP Server - Expose

Primitives:

Tools: tools/list

Resources: resources/list

Prompt Templates: prompts/list

The Protocol - resources

resources/list

List all resources

```
{  
  "jsonrpc": "2.0",  
  "id": 7,  
  "method": "resources/list"  
}
```

resources/read

Read a resource

```
{ "jsonrpc": "2.0", "id": 1, "method": "resources/read",  
  "params": { "uri": "meeting://notes/2024-09-15" } }
```

resources/subscribe

Subscribe to resource
changes

```
{ "jsonrpc": "2.0", "id": 3, "method": "resources/subscribe",  
  "params": { "uri": "calendar://events/today" } }
```

The Protocol - tools

tools/list

List all tools

Request

```
{  
  "jsonrpc": "2.0",  
  "id": 9,  
  "method": "tools/list"  
}
```

Response

```
{  
  "jsonrpc": "2.0",  
  "id": 9,  
  "result": {  
    "tools": [  
      {  
        "name": "search_meeting_notes",  
        "description": "Search meeting notes by date and project.",  
        "inputSchema": {  
          "type": "object",  
          "properties": {  
            "date": { "type": "string", "description": "ISO date, e.g. 2024-09-15" },  
            "project": { "type": "string" }  
          },  
          "required": ["date"]  
        }  
      },  
      {  
        "name": "update_task",
```

The Protocol - tools

tools/call

Use a tool

```
{  
  "jsonrpc": "2.0",  
  "id": 10,  
  "method": "tools/call",  
  "params": {  
    "name": "search_meeting_notes",  
    "arguments": {  
      "date": "2024-09-15",  
      "project": "launch"  
    }  
  }  
}
```


Lifecycle	notifications/initialized	initialize, ping
Tools	notifications/tools/list_changed	tools/list,★ tools/call ★
Resources	notifications/resources/updated, notifications/resources/list_changed	resources/list,★★ resources/read, resources/templates/list, resources/subscribe, resources/unsubscribe
Prompts	notifications/prompts/list_changed	prompts/list,★ prompts/get ★
Client features	notifications/roots/list_changed	sampling/createMessage, elicitation/create, roots/list, completion/complete
Utilities	notifications/message, notifications/progress, notifications/cancelled	logging/setLevel

<https://modelcontextprotocol.io/>

Benefits of Model Context Protocol (MCP)

Open & Interoperable

MCP is an open standard, meaning anyone can build on it, no vendor lock-in.

Two-Way Communication

Enables AI tools to both read and write to external systems securely and reliably.

Smarter AI

By giving LLMs access to real-time, structured, and contextual data, you make agents more useful, accurate, and grounded.

Privacy & Security Built-In

Designed with permissions, authentication, and secure data handling in mind.

Plug-and-Play Integrations

Supports a growing ecosystem of ready-to-use connectors for tools like Notion, Slack, Google Drive, APIs, databases, and more – no need to build from scratch.

<https://github.com/modelcontextprotocol/servers>

Project Presentations

Project presentations

Dates:

Batch 1: **6.7.2026**

Batch 2: **13.7.2026**

Batch 3: **20.7.2026**

Bonus:

The best and second-best presentations in each batch will receive **+3 and +2 bonus points**, respectively, added to the final score.

Questions?